

Amazon ML Challenge 2026: Multi-Source Business Entity Resolution

Submission #5 Technical Documentation & Methodology Report

Team Name: Antigravity ER

Submission Identifier: Rahul_AmazonML_Submission5

Public Leaderboard Score: 0.848812

Pipeline Version: V13.1 Two-Stage Collective Matcher + conservative singleton recovery

Validation: V13.1 core: Macro $F_{(0.5)} = \mathbf{0.90143}$ across four held-out splits; #5 recovery ablation: 0.89516 baseline to 0.89617 for the selected policy. These are offline results, not the public score.

Submission Validator Status: PASS for required matching-file format and row checks; optional candidate containment was not part of this validation run.

1. Executive Summary

Submission #5 keeps Submission #3 as its frozen prediction core and adds only 7,847 high-confidence empty-to-singleton recoveries. The rescue gate is rank 1, score at least 0.85, and top-1/top-2 margin at least 0.15; it added no links to already non-empty S1 rows and removed no existing links. The public leaderboard score recorded for #5 is 0.848812. The score is not 0.91+, and the offline figures below must not be represented as leaderboard performance.

The underlying Submission #3 matcher is the **V13.1 Two-Stage Collective LightGBM Matcher** with $\alpha = 0.50$, bounded multi-view retrieval, R50 rescoring, collective sibling evidence, and target exclusivity. Its four-split offline result was reported as 0.90143. The separate #5 recovery ablation reports 0.89516 for its baseline and 0.89617 for the selected recovery policy; use these only with their respective evaluation context.

2. Methodology

2.1 Entity Resolution Formulation

The challenge requires matching records from Source 1 (S1) to corresponding real-world entities in Source 2 (S2) and Source 3 (S3):

1-to-many S1 mapping: A single S1 entity can match multiple target records (branch locations, multi-lingual aliases, legal entity siblings).

Target Exclusivity: A single target record in S2 or S3 can belong to at most one true business entity in S1.

Evaluation Metric: Macro-averaged $F_{(0.5)}$, placing 4x more weight on precision than recall ($\beta = 0.5$), making false positive avoidance and singleton discrimination critical.

2.2 Preprocessing & Unicode Normalization

Unicode Combining Mark Preservation: Used `[^p{L}\p{M}\p{N}\s]` regular expressions to prevent separating Indian vowel matras from base consonants across Hindi, Telugu, Tamil, and Kannada scripts.

Diacritical Stripping: Stripped Latin accents (e.g. `coalition` to `coalition`) while keeping Indic scripts intact.

Legal Entity & Web Artifacts: Normalization of company suffixes (pvt ltd, inc, llc, gmbh) and stripping of domain suffixes (.com, .net, .org).

2.3 Candidate Generation (Blocking) & V13 Bounded Sparse Retrieval

To balance recall and candidate volume:

- Multi-Pass Core Blocking:** Compact unaccented name, rare name tokens (DF ≤ 150), normalized address, street number + street token, and rare address tokens (DF ≤ 40).
- V13 Bounded Sparse Retrieval:**
 - View A:** Mark-safe Unicode normalization retaining dual native-script and Latin representations.
 - View B:** Learned Indic-to-Latin transliteration dictionary mined strictly on `train_gate` (1,322 Source 2 tokens, 1,135 Source 3 tokens; zero test leakage).
 - View C:** Fast OCR-tolerant prefix retrieval mapping common character confusions ('1' $\rightarrow$ 'l', '0' $\rightarrow$ 'o').
 - View D:** Address token joins within country partitions.

2.4 True Frozen R50 Rescoring

All candidate pairs are enriched with the frozen R50 feature vector and evaluated directly through `models/lgb_source2.txt` and `models/lgb_source3.txt`. This eliminates missing-feature degradation and recovers high base scores (mean 0.538, 83.0% > 0.1) on newly retrieved candidates.

2.5 Two-Stage Collective Matcher & Sibling Evidence

- Stage-1 LightGBM:** Trained with 41 deterministic pairwise string, token, edit-distance, address, and competition features.
- Entity-Level 3-Fold Out-Of-Fold (OOF) Prediction:** Generated unbiased Stage-1 probabilities across training entities with zero entity leakage.
- Collective Sibling Features:** Extracted 12 collective signals capturing sibling consensus:
 - High-confidence sibling count (`s1_sibling_high_conf_count`)
 - Maximum sibling score and margin (`s1_best_sibling_score` , `s1_sibling_margin`)
 - Target cluster consensus agreement (`cluster_s1_agreement` , `cluster_max_score`)
 - Reverse target competition (`target_stage1_margin` , `target_is_stage1_best`)
- Stage-2 LightGBM:** Trained on 69 combined features to model multi-entity sibling consensus.

2.6 Alpha Blend Ensemble & Target Exclusivity

Ensemble Blend: $\text{Final Score} = 0.50 \cdot \text{Stage-1} + 0.50 \cdot \text{Stage-2}$.

Selected Thresholds:

Source 2: $\tau = 0.44$

Source 3: $\tau = 0.47$

Global Target Exclusivity: Enforces that each target record $\$T\$$ is assigned to at most one $\$S1\$$:

$$S_1^*(T) = \arg\max_{S_1} \left(\text{Final Score}(S_1, T) \right)$$

Pruned 318,915 ambiguous collision links, completely eliminating target conflicts (0 violations).

3. Validation Results

The table below reports the underlying V13.1 core before Submission #5's test-only singleton recovery. It is offline validation on `val1` and `val2`, not a leaderboard result. The separate #5 recovery ablation reports `0.89516` for its baseline and `0.89617` for the selected `score ≥ 0.85`, `margin ≥ 0.15`, `rank-1` policy.

[illegible]

4. Final Test Statistics

Complete test inference was executed across the full test candidate graph:

S1 rows (matching_results.tsv): 1,732,544

S1 rows (candidate_pairs.tsv): 1,732,544

S2 links: 2,231,034

S3 links: 2,400,847

Total predicted links: 4,631,881

Empty S1 entities: 161,929

Non-empty S1 entities: 1,570,615

#3-to-#5 changes: 7,847 entities, all empty-to-non-empty; 7,847 links added, zero removed.

Singleton recoveries preserved: 7,847 / 7,847 in the frozen #5 matching artifact.

Candidate containment: The #5 candidate file is byte-identical to #3's. The official validator's optional large candidate cross-check was not run in the final matching-only check.

Validator: Matching-file required row and format checks passed; candidate containment and optional ID-existence checks remain unverified here.

5. Artifact Hashes & Integrity

matching_results.tsv SHA-256: `b3df36d5ff924d7cf7e0f0040a69dc484257ce69196545308c2288bb53d9e87c`

candidate_pairs.tsv SHA-256: 8452fd1f0b586ff1d1374fef108273a24ab30f6782043a297f2f65a4ee7e39f6

models/lgb_source2.txt SHA-256: `ecabacfb6acb9b567d04956928b5b3011e9951a0bb837441bede0e512b2aee9` (VERIFIED UNTOUCHED)

models/lgb_source3.txt SHA-256: 304aa86a6cd1bdccb025ffeefc852958fa7c3b5d165ced2bdc42d3429179f2f40 (VERIFIED UNTOUCHED)

models/v13_1_stage1_source2.txt SHA-256: 1998672fdae6f094f2ccbf965e8f23517344638ce6598ed29ff9adc5169af67d

models/v13_1_stage2_source2.txt SHA-256: 354d66b4ca47636b8b70200321b158f75ae5f23caaaa87170f623a71f2601018

models/v13_1_stage1_source3.txt SHA-256: 7436fdd685e6ec143e3d90cb99908ba2aa8be8580618396470831009016bc98f

models/v13_1_stage2_source3.txt SHA-256: 8f4c5ad5ed4480654b6622b93cedd7aa5c18eef6b61ca692382a40f0d177dceb